49. What is the purpose of setting up of Small Finance Banks (SFBs) in India?

1. To supply credit to small business units
2. To supply credit to small marginal farmers
3. To encourage young entrepreneurs to set up business particularly in rural areas.

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (a) Small Finance banks are a type of niche bank in India. Banks with a small finance bank license can provide basic banking services of acceptance of deposit and lending. The objective of setting up small finance banks is to further financial inclusion by the provision of savings vehicles and supply credits to small business units, small and marginal farmers, micro and small industries and other unorganized sector entities through high technology-low cost operations, However apart from the provision of credit there is no explicit provision that SFBs have been setup to encourage setting of business in rural areas. Hence, statement 3 is incorrect.

• Foreign Trade & Economic Organization of India, Stock Market

50. 'Broad-Based Trade and Investment Agreement (BTIA)' is sometimes seen in the news in the context of negotiations held between India and

(a) European Union
(b) Gulf Cooperation Council
(c) Organization for Economic Cooperation and Development
(d) Shanghai Cooperation Organization

Ans : (a) Negotiations for a free trade pact, formally called a broad- based trade and investment agreement (BTIA), between the EU and India, started in 2007, but were unofficially suspended in 2013 when both sides realized that no progress was happening on contentious matters.
EU has raised concern over what is called India's "unilateral termination" of existing Bilateral Investment Treaties (BITs) with a significant number of "EU member countries."

51. Consider the following statements:

1. India has ratified Trade Facilitation Agreement (TFA) of WTO.
2. TFA is a part of WTO's Bali Ministerial Package of 2013.
3. TFA came into force in January 2016.

Which of the statement given above is/are correct?

(a) 1 and 2 only (b) 1 and 3 only
(c) 2 and 3 only (d) 1, 2 and 3

Ans : (a) WTO members concluded negotiations at the 2013 Bali Ministerial conference on the landmark Trade Facilitation Agreement (TFA), which entered into force on February 22, 2017 following its ratification by two-thirds of the WTO membership. India has ratified the Trade Facilitation Agreement (TFA) of the World Trade Agreement (WTO) and the instrument of acceptance for Trade Facilitation Agreement was handed over to WTO Director- General by India on April 26, 2016.

52. The Global Infrastructure Facility is a/an

(a) ASEAN initiative to upgrade infrastructure in Asia and financed by credit from the Asian Development Bank.
(b) World Bank collaboration that facilitates the preparation and structuring of complex infrastructure Public–Private Partnerships (PPPs) to enable mobilization of private sector and institutional investor capital.
(c) Collaboration among the major banks of the world working with the OECD and focused on expanding the set of infrastructure projects that have the potential to mobilize private investment.
(d) UNCTAD funded initiative that seeks to finance and facilitate infrastructure development in the world.

Ans : (b) The Global Infrastructure Facility (GIF) was launched by the World Bank to cater to the Infrastructure needs of countries. It is a global open platform that facilitates the preparation and structuring of complex infrastructure public- private partnerships (PPPs) to enable mobilization of the private sector and instructional investor capital. The GIF platform coordinates and integrates the efforts of Multilateral Development Banks (MDBs), private sector investors and financiers and governments interested in infrastructure investment in Emerging Markets and Developing Economies (EMDEs).

SCIENCE

• Physics

53. What is the purpose of 'evolved Laser Interferometer Space Antenna (eLISA)' project?

(a) To detect neutrinos
(b) To detect gravitational waves
(c) To detect the effectiveness of missile defence system
(d) To study the effect of solar flares on our communication systems

Ans : (b) The Laser Interferometer Space Antenna (Spectacular plan of setting into space three Spacecraft) is a European Space Agency mission designed to detect and accurately measure gravitational waves tiny ripples in the fabric of space time-from astronomical sources. LISA would be the first dedicated space-based gravitational wave detector.

• Chemistry

54. Consider the following pairs:

Commonly used/consumed materials		Unwanted or controversial chemicals likely to be found in them
1. Lipstick	–	Lead
2. Soft drinks	–	Brominated vegetable oils
3. Chinese fast food	–	Monosodium glutamate

Which of the pairs given above is/are correctly matched?

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) Lead is a common impurity found in cosmetic lip products, Such as lipsticks and other cosmetics, such as eye shadows, blushes, compact powders, shampoos and body lotions.
Brominated Vegetable oil is a food additive sometimes used to keep citrus flavouring from separating out in the some sodas and soft drinks. It is banned in Europe against use as a food additive in Europe and Japan.
Monosodium Glutamate (MSG) is a Glutamate or salt of Glutamic acid, a "non-essential" amino acid. It is a neurotransmitter- transporting messages from one nerve cell to another. Glutamic acid is found naturally in tomatoes, grapes, cheese, mushrooms and other foods. It is commonly used in Chinese fast food.

55. Organic Light Emitting Diodes (OLEDs) are used to create digital display in many devices. What are the advantages of OLED displays over Liquid Crystal displays?

1. OLED displays can be fabricated on flexible plastic substrates.
2. Roll–up displays embedded in clothing can be made using OLEDs.
3. Transparent displays are possible using OLEDs

Select the correct answer using the code given below:

(a) 1 and 3 only
(b) 2 only
(c) 1, 2 and 3
(d) None of the above statements is correct

Ans : (c) A major advantage of OLED over LCD is that it is lightweight and has flexible plastic substrates. OLED displays can be fabricated on flexible plastic Substrates leading to the possible fabrication of flexible organic light-emitting diodes for other new applications, such as roll-up displays embedded in fabrics or clothing. If a substrate like polyethylene terephthalate (PET) can be used, the displays may be produced inexpensively; furthermore, plastic substrates are shatter- resistant unlike the glass displays used in LCD devices.

• Medical Science, Zoology, Anthropology

56. What is the application of Somatic Cell Nuclear Transfer Technology?

(a) Production of biolarvicides
(b) Manufacture of biodegradable plastics
(c) Reproductive cloning of animals
(d) Production of organisms free of diseases

Ans : (c) In genetics and developmental biology, Somatic Cell Nuclear Transfer (SCNT) is a laboratory strategy for creating a viable embryo from a body cell and an egg cell. The technique consists of taking an enucleated oocyte (egg cell) and implanting a donor nucleus from a somatic (body) cell. It is used in both therapeutic and reproductive cloning. Dolly the sheep became famous for being the first successful case of the reproductive cloning of a mammal.

57. Consider the following statements:

1. In tropical regions, Zika virus disease is transmitted by the same mosquito that transmits dengue.
2. Sexual transmission of Zika virus disease is possible.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 only
(c) Both 1 and 2 (d) Neither 1 nor 2

Ans : (c) Statement 1 : Zika Virus is transmitted to people primarily through the bite of an infected Aedes species mosquito (Ae. aegypti and Ae. albopictus). These are the same mosquitoes that spread dengue and chikunguniya viruses. Hence, it is correct.
Statement 2: Zika can be passed through sex from a person who has Zika to his or her partners. Zika can be passed through sex even if the infected person does not have symptoms at the time. Hence, it is correct.

• Agriculture, Animal Husbandry & Dairy

58. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host–pathogen relationships in crops.

Select the correct answer using the code given below:

(a) 1 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans : (d) All the statements are correct. Genome Sequencing is figuring out the orders of DNA nucleotides or bases in a genome-the order of As, Cs, Gs, and Ts that make up an organism's DNA. Genome Sequencing of wide varities of plants can be used to identify disease resistance and drought tolerance genes in various plants and develop new varieties of crop plants in lesser time. Genome sequencing of crop plants can be helpful in deciphering and understanding the host- pathogen relationship in crops.